

HFD 207: TEXTILE AND DESIGN APPLICATION

PRACTICAL HANDOUT: TIE-DYEING USING VAT DYES

Practical: Tie-Dyeing with Vat Dyes

Fabric: 100% Cotton

Reference Fabric Weight: 100 g

Method: Tie-and-dye resist technique

1. Practical Objectives

By the end of the practical, the student should be able to:

1. Prepare cotton fabric for vat dyeing.
2. Create a tie-dye resist pattern using folding, tying or knotting.
3. Apply vat dye to cotton fabric.
4. Understand the processes of reduction, dyeing and oxidation.
5. Wash and finish the dyed fabric appropriately.

2. Materials and Equipment

Materials/Equipment	Purpose
White 100% cotton fabric – approximately 100 g	Dyeing material
Vat dye	Colouring agent
String/thread/rubber bands	Creating resist areas
Plastic basin/container	Dye bath
Stirring stick	Mixing
Measuring spoon	Measuring non-hazardous materials where appropriate
Gloves and protective clothing	Personal protection
Caustic soda solution	Alkaline medium
Sodium hydrosulphite solution	Reducing agent
Water	Dye bath and washing
Soap/detergent	After-treatment

3. Fabric Preparation

1. Cut approximately **100 g of clean cotton fabric**.
2. Wash/scour the fabric thoroughly to remove dirt, oil and other finishes.
3. Rinse well.
4. Keep the fabric damp.
5. Fold, twist, pleat, knot or scrunch the fabric according to the selected tie-dye design.
6. Tie firmly using string, thread or rubber bands.

4. Dye-Bath Procedure

The following sequence should be followed:

Vat dye + water → reduction with prepared alkaline/reducing solution → dye bath → immerse tied fabric → oxidation → washing → soaping → final rinse

The amount of vat dye should be determined according to the required shade and the manufacturer's recommended percentage of the fabric weight.

For the purposes of this class we will use:

Fabric Quantity	Dyebath quantity	Vat Dye	Caustic Soda	Sodium Hydrosulphite
3 metres	9 litres of water	1 table spoon	1 tablespoon	1 ½ table spoon
1 metre	2 litres of water	½ table spoon	½ tablespoon	1 tablespoon

5. Dyeing Procedure

1. Prepare the vat dye bath under the supervision of the lecturer/technician.
2. Ensure that the tied cotton is completely wet before entering the dye bath.
3. Immerse the tied fabric carefully.
4. Move and turn the fabric gently to encourage dye penetration.
5. Maintain the recommended dyeing temperature and time for the particular vat dye.
6. Remove the fabric from the bath.
7. Allow it to oxidise in air. The colour develops as the reduced dye is converted back to its insoluble coloured form.
8. Rinse thoroughly.
9. Untie the fabric carefully to reveal the pattern.
10. Rinse again.
11. Soap/wash the fabric according to the dye manufacturer's after-treatment recommendation.
12. Rinse, dry and press.

7. Safety

- Wear gloves and protective clothing.
- Work in a well-ventilated area.
- **Never handle concentrated caustic soda or sodium hydrosulphite with bare hands.**
- Do not mix chemicals casually or change the prescribed recipe.
- Wash hands and equipment thoroughly after the practical.

Expected Outcome

A successfully dyed cotton fabric showing a clearly defined **tie-dye resist pattern**, good colour development and adequate washing/soaping.

Lecturer: Paul Masele Litaba

Course: HFD 207 – Textile and Design Application